

Experiment 40: Draw Rectangular Shape and Extract Objects

Aim

To draw rectangular bounding boxes around objects in an image and extract the selected objects using OpenCV.

Procedure

1. Import the OpenCV library.
2. Read the input image.
3. Specify the rectangular coordinates of the objects.
4. Draw rectangles around the selected objects.
5. Extract the objects using image slicing.
6. Display the original image with rectangles.
7. Display the extracted objects.
8. Save the extracted objects if required.

Program

```
import cv2

# Read the input image

img = cv2.imread("input.jpg")

if img is None:

    print("Image not found")

    exit()

# Rectangle for first object

x1, y1, w1, h1 = 90, 15, 85, 90
```

```
# Rectangle for second object
```

```
x2, y2, w2, h2 = 170, 15, 75, 100
```

```
# Draw rectangles
```

```
cv2.rectangle(img, (x1, y1), (x1+w1, y1+h1), (0, 255, 0), 2)
```

```
cv2.rectangle(img, (x2, y2), (x2+w2, y2+h2), (255, 0, 0), 2)
```

```
# Extract objects
```

```
object1 = img[y1:y1+h1, x1:x1+w1]
```

```
object2 = img[y2:y2+h2, x2:x2+w2]
```

```
# Display original image with rectangles
```

```
cv2.imshow("Rectangular Objects", img)
```

```
# Display extracted objects
```

```
cv2.imshow("Extracted Object 1", object1)
```

```
cv2.imshow("Extracted Object 2", object2)
```

```
# Save extracted objects
```

```
cv2.imwrite("object1.jpg", object1)
```

```
cv2.imwrite("object2.jpg", object2)
```

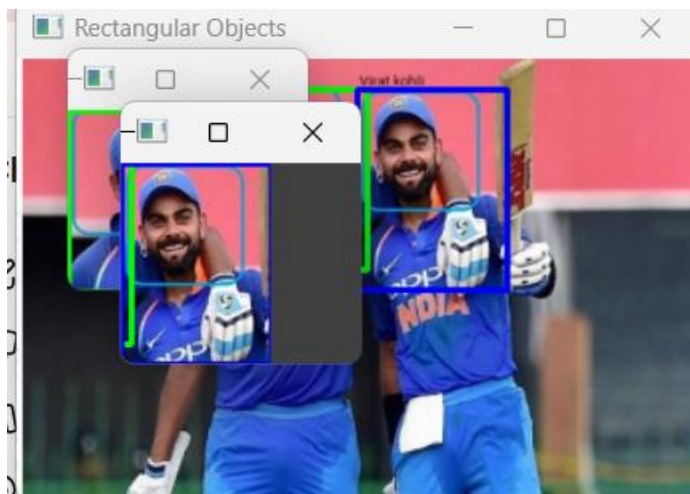
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the rectangular shapes were successfully drawn around the selected objects and the objects were extracted from the given image using OpenCV.